

Chandan Singh

Computer Science Undergraduate | Software Engineering Intern Candidate

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Professional Summary

Third-year Computer Science Engineering student with hands-on experience in full-stack development, relational databases, and applied machine learning. Proficient in Python, C++, Java, SQL, and JavaScript, with growing expertise in FastAPI, LangChain/LangGraph, and RAG pipelines. Built and deployed four end-to-end projects - from responsive web interfaces and ML-based fraud detection to a production-style AI chatbot application.

Technical Skills

Languages: C++, Java, Python, SQL

Web Technologies: HTML5, CSS3, JavaScript (ES6+)

Frameworks & Libraries: Flask, FastAPI, LangChain, LangGraph, XGBoost

Databases & Tools: Oracle DB, SQLite, ChromaDB, Git, GitHub, VS Code, Google Sheets API

Core Concepts: DSA, OOP, DBMS, Operating Systems, Machine Learning, RAG Pipelines, JWT Authentication

Projects

Voxa AI – Chatbot Application | *FastAPI, LangGraph, LangChain, RAG Pipelines* | Live · GitHub 2026

- Architected a 7-microservice backend (auth, RAG, agent, tools, DB, frontend, guardrails) with a 7-model LLM fallback chain across 2 providers (Groq + OpenRouter) and JWT-based authentication with bcrypt password hashing.
- Built a RAG pipeline using ChromaDB vector store with HuggingFace embeddings, delivering SSE-based response streaming with sub-2-second average server latency (TTFB) and a 99/100 Google Lighthouse performance score; Dockerized and deployed on Hugging Face Spaces.

Personal Portfolio Website | *HTML5, CSS3, JavaScript* | Live · GitHub 2026

- Designed and developed a fully responsive portfolio site using semantic HTML, CSS Flexbox/Grid, and responsive design principles, ensuring consistent rendering across Chrome, Firefox, and Safari.
- Optimized performance using Google Lighthouse audit methodology, achieving a 93/100 performance score; site is live and publicly accessible for recruiter outreach.

Student Management System | *Python-Flask, SQL, Oracle DB* | GitHub 2025

- Built a full-stack CRUD system on a normalized Oracle SQL backend with primary/foreign key constraints, managing 500+ student records with sub-second query response times.
- Implemented secure session-based authentication, preventing unauthorized access to administrative functions in all tested scenarios.

Real-Time Fraud Detection System | *Python-Flask, XGBoost, Oracle DB* 2024

- Trained an XGBoost classification model on labelled transaction data, achieving 80% classification accuracy on a balanced test set of 100+ transactions.
- Eliminated manual entry for 100% of flagged records by automating an end-to-end data ingestion pipeline using Google Sheets API into an Oracle DB audit layer.

Education

Chandigarh University

B.E. in Computer Science and Engineering — CGPA: 8.04 / 10.0

Punjab, India

2024 – 2028

Academic Achievements

- Selected for TES (52) entry and appeared before the Services Selection Board (SSB).
- Awarded a 40% scholarship at Chandigarh University in recognition of academic excellence.
- Scored 90 percentile in the Joint Entrance Examination (JEE).

Certifications

- NPTEL (Elite): Introduction to Machine Learning by IIT Madras
- LinkedIn Learning: Become a Full-Stack Web Developer
- Coursera / IBM: Introduction to Software Engineering
- Coursera / University of Michigan: Using Databases with Python
- Coursera: Object-Oriented Programming and Functions